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An Interval Multi-objective Evolutionary Generation Algorithm for Product Design Change Plans in Uncertain Environments

Rui-zhao Zheng, Yong Zhang, Xiao-yan Sun, Dun-wei Gong, Xiao-zhi Gao

***Abstract*—Design change is an important issue in complex product development projects. In a complex product with numerous parts (also known as components), the change of one key part may spread to other parts associated with it, generating a chain reaction throughout the entire project. Therefore, it is necessary to select a suitable change plan involving only fewer crucial parts in order to enhance the product's performance, minimize change cost, and reduce change duration/time. Focusing on the case where the correlation strength between parts cannot be accurately obtained, in this paper we study an interval multi-objective evolutionary algorithm for finding excellent design change plans. Firstly, on the basis of the established multi-layer product network with interval correlation weights, an interval multi-objective optimization model of the product design change planning problem is established, where three new objective functions regarding product performance, carbon trading cost and supply risk are defined. Then, a constraint multi-objective evolutionary algorithm based on interval Pareto dominance is proposed to search for optimal change plans. Several novel operators, including the problem characteristic-guided population update strategy, the probability-based interval Pareto dominance, and the interval constraint handling strategy, are developed to enhance the algorithm's performance. Finally, the proposed algorithm is compared with eight existing algorithms on the two design change cases, experimental results revealed its effectiveness.**

***Index Terms*—Product change; evolutionary optimization; multi-objective; uncertainty**

I. Introduction

In real-life production, products often need to be re-designed due to many reasons, including customer demand change, product upgrading, and supply chain disruptions [1], [2]. Product design changes are inevitable during product development, and the change cost usually reaches 20% to 50% of the total design cost [3], [4]. Due to the complex coupling relationship between product parts, changes to certain parts often impact other related parts [5]. Without effective control, these design changes can rapidly propagate through the parts [6], subsequently impacting product performance, cost, and duration, and can even lead to change failures [7], [8], [9]. Consequently, studying the issue of product design changes is of great significance.

Many methods have been proposed to address design change problems up to now. Most existing methods employ complex network theory to deal with the kind of problem, where product parts are abstracted as nodes, and relationships between parts are used as edges of product network [10], [11]. Yin et al. [12] transformed mechanical product parts and their dependencies into a directed network, and presented a change propagation path optimization method based on Dijkstra. Qiao et al. [13] proposed a hybrid method based on multi-dimensional scaling and clustering for constructing design structure matrices (DSMs), enhancing various product attributes at different life cycle stages. Liu et al. [14] proposed a change impact evaluation method based on breadth-first search, considering the resistance of part design tolerances to changes.

Many-objective optimization approaches have recently been used to solve the product design and supply chain robustness design/redesign problem. Singh et al. [15] introduced an automated redesign method, wherein a decomposition-based many-objective evolutionary algorithm is used to find a trade-off set of design variants that spans a range of feasibility and performance robustness. Karadag et al. [16] presented a multi-objective location-allocation model for the blood supply chain design problem. By minimizing the distances between the supply chain elements and the length of the mobile unit routes, the model can indirectly obtain an optimal plan with reduced transport time and cost. Luo et al. [17] introduced an NSGA-II-based change propagation path optimization approach, and verified its effectiveness with a calculator design problem. Considering essential indicators such as part modularity and design change propagation scope, Li et al. [18] developed a design change propagation path optimization model and a multi-population genetic algorithm for solving the model. Ullah et al. [19] considered overall change propagation risk, and proposed a method for predicting the minimum risk change propagation path in product structures. These methods provide various effective modeling and optimization approaches for solving complex design change problems, but they did not consider the uncertain information in the product modeling process, particularly the uncertainty of correlation weights between parts.

It is widely recognized that due to the lack of sufficient

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historical change data and the subjective characteristics of experts, it is difficult for decision-makers to accurately evaluate the degree of correlation between parts when modeling complex products, and can only provide their uncertain or approximate values [20]. In view of this, in recent years, scholars have begun to study supply chain congestion or interruption problems with uncertain parameters. Some representative achievements in this area include the enhanced possibilistic programming method for reliable closed-loop supply chain network design [21], the multi-objective uncertain equilibrium model for green supply chain networks [22], the fuzzy-stochastic multi-objective mathematical model for closed-loop supply chain networks [23], and the closed-loop supply chain model integrating customer demand response speed [24], and so on. These methods have proven helpful in managing uncertainties; however, they are all explicitly designed for supply chain-related issues. Given that the change propagation model mode among product parts significantly differs from the information dissemination patterns among suppliers, these results may not be directly applicable to product design change problems in uncertain environments.

In view of it, this paper studies an interval multi-objective evolutionary algorithm for uncertain product design change plan problems, in which the correlation strength (i.e., associated weights) between parts represented as interval numbers, referred to as interval weights. On the basis of the established product design multi-layer network model with interval weights, an interval multi-objective optimization model for product design change plans is first established, and then an interval multi-objective evolutionary algorithm is presented. The main contributions of this paper are as follows:

(1) Constructing a product multi-layer network model with interval weights for product design change problems. This model first discusses the interval uncertainty of associated weights between network nodes while introducing product performance and supplier information. As a result, the proposed model is more closely aligned with the practical requirements of product design problems.

(2) Establishing an interval multi-objective optimization model for product design change plans. Three new indicators, including product performance impact degree, carbon trading cost, and supply risk, are defined to comprehensively evaluate a design change plan under interval uncertainty.

(3) Proposing a constrained multi-objective evolutionary algorithm based on interval Pareto dominance criterion to generate a high-quality set of change plans. Several new operators, including the problem characteristic-guided population update strategy, the probability-based interval Pareto dominance criterion, and an interval constraint processing strategy, significantly enhance the algorithm's performance.

The structure of this paper is as follows: Section II establishes the product multi-layer network model; Section III presents the interval multi-objective optimization model for product design change plans; Section IV introduces the proposed multi-objective evolutionary algorithm; Section V experimentally validates the proposed algorithm's effectiveness; Section VI concludes the paper.

II. Product Multi-layer Network Model with Interval Weights

A complex product often consists of many parts with intricate linkages [25]. Most existing studies primarily focus on the linkage relationships between parts to construct single-layer product design network models [26]. Although a few researchers have introduced product performance to evaluate the impact of a change plan on user satisfaction, and added supplier information to study multi-layer network models of complex products [27], these methods do not consider uncertain information in the product design process. As a result, the constructed network models are not easily applicable to real-world product design problems in uncertain environments. To address this issue, this section introduces the concept of interval weight to present the uncertainty in associated weights between parts or nodes. Interval weight is a unique associated weight represented as interval numbers. Based on this, we then propose a product multi-layer network model with interval weights.

A. Product Multi-layer Network Model

This section transforms complex product design problems into a 3-layer network model to analyze the implication of product design changes on suppliers and service performance. For a product PR containing n parts and m service performances, as shown in Figure 1, the corresponding 3-layer network model is specified as follows:

$$\begin{aligned} &PR = \{SP, P, S\} \\ &SP = \{sp_1, sp_2, \cdots, sp_k, \cdots sp_m\} \\ &P = \{p_i|_{i=1,2,\ldots,n}, e_{ij}|_{i,j=1,2,\ldots,n,i\neq j}, w_{ij}|_{i,j=1,2,\ldots,n,i\neq j}\} \\ &S = \left\{ \begin{array}{l} (s_1^0, s_1^1, \ldots s_1^{|s_1|}), \cdots, (s_i^0, s_i^1, \ldots s_i^{|s_i|}), \cdots, \\ (s_n^0, s_n^1, \ldots s_n^{|s_n|}) \end{array} \right\} \end{aligned} \tag{1}$$

In the above network, the service performance layer SP contains multiple service performances that the product must meet. The part network layer P uses a complex network to describe the attributes of parts and the correlation between parts, where $\{p_i|i = 1, 2, \cdots, n\}$ is the set of nodes, $\{e_{ij}|i, j = 1, 2, \cdots, n, i \neq j\}$ represents the set of edges, and $\{w_{ij}|i, j = 1, 2, \cdots, n, i \neq j\}$ represents the set of weights. The supplier layer S records the product's supplier information. $(s_i^0, s_i^1, \ldots s_i^{|s_i|})$ represents the set of suppliers for the $i-\text{th}$ part p_i, where $|s_i|$ represents the number of alternative suppliers for the part p_i, and s_i^j represents the $j-\text{th}$ supplier for the part p_i.

1) Parts Network Layer P: Complex products are composed of numerous parts. The part network layer primarily records the attributes of parts and the correlation between parts. Part attributes encompass cost, duration, carbon emissions, parameter performance, and so on. A part's duration refers to the time required for the re-development of a new part or the difference in production and procurement time between the old part and its replacement. The parameter performance of a part is mainly used to evaluate the impact of the parameter values of the part on the product performance. In other words, the parameter performance of a part can be evaluated by

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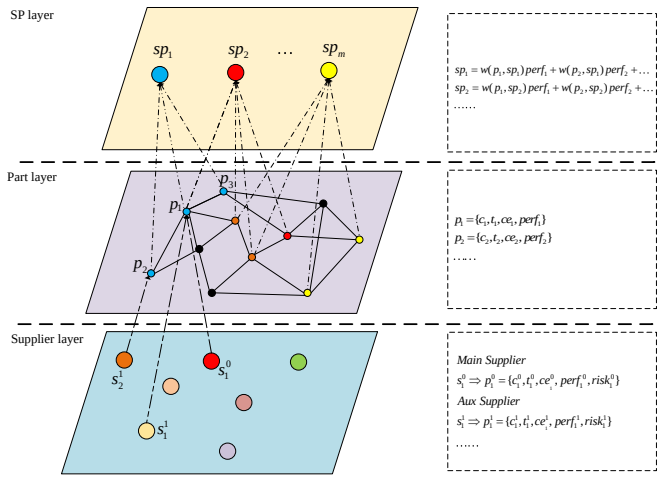


Fig. 1. The Constructed Product Multi-layer Network Model

analyzing the impact of its key parameters' changes on the product performance. As we all know, the performance of a product is determined by the performance parameters of all its components/parts. For example, the parameter performance of the part television screen can be evaluated based on parameters such as color, resolution, size, etc., while the parameter performance of the part speaker needs to consider parameters such as power, impedance, frequency response.

To maintain generality, for a part p_i within the product, we use attributes such as cost c_i, duration t_i, carbon emissions ce_i, and parameter performance perf_i to characterize it. These attributes can be denoted as:

$$p_i = \{\mathrm{c}_i, \mathrm{t}_i, \mathrm{ce}_i, \mathrm{perf}_i\} \tag{2}$$

There is a coupling relationship between parts within a product, so changes in one part may impact other related parts. To capture this mutual influence, an associated weight between part p_i and part p_j, denoted as w_{ij}, is used [28]. Specifically, if $w_{ij} = 0$, it signifies that there is no connection between part p_i and part p_j, indicating that changes in the attributes of part p_i do not affect part p_j, or the impact is negligible.

As we all know, accurately determining the associated weights between parts is crucial for modeling the complex product network. Up to now, scholars have proposed a lot of effective methods to determine the associated weights between parts. However, these methods either require a large amount of data or precise expert experience. In many practical applications, engineers can only obtain a small amount of data, and the empirical values they provide also have strong subjectivity. To address this issue, we use an interval value to describe the associated weight between two parts. Specifically, the associated weight between parts p_i and p_j is $\bar{w}_{ij} = [{w_{ij}}^-, {w_{ij}}^+],\ 0 \le {w_{ij}}^- \le {w_{ij}}^+ \le 1$.

2) Service Performance Layer SP: Generally, it is difficult to obtain user satisfaction during the product design stage directly; however, the parameter performance of relevant parts in the product can indirectly estimate the satisfaction level of users with a certain service performance [27]. For example, the display quality of a TV is mainly determined by the parameter performance of the screen panel part, while the TV's wireless service performance is mainly determined by parts such as wireless network cards and antennas. Service performance is an indirect indicator that evaluates the extent to which a product meets specific user satisfaction [27]. Noted that, since a part may be associated with multiple service performances, there is a correlation relationship between service performances. In this paper, the influence strength of part p_i on the service performance sp_k is denoted as iw_{ik}. Specifically, if $iw_{ik} = 0$, it means that the part p_i does not impact the service performance sp_k.

3) Supplier Layer S: In general, a part of a product has multiple suppliers. For the part p_i, let its supplier set be $(s_i^0, s_i^1, \ldots s_i^k)$. Here, s_i^0 is its primary supplier, that is, the initial supplier when the product has not undergone any changes, and others are its alternative suppliers. When the initial supplier exits due to issues, a replacement can be selected from these alternative suppliers. Different suppliers have different supply costs and supply risks, they provide parts with different attribute values. Based on their historical reputation and supply stability, the supply risk (denoted as $risk$) of a supplier when it supplies a part can be determined. This paper assumes that supply risk is known. For the part p_i in the product, the deliverable provided by the initial supplier s_i^0 (i.e., the part without changes) and its attributes are denoted as $p_i^0 = \{\mathrm{c}_i^0, \mathrm{t}_i^0, \mathrm{ce}_i^0, \mathrm{perf}_i^0, \mathrm{risk}_i^0\}$. The deliverable provided by the j-th alternative supplier and its attributes are denoted as: $p_i^j = \{\mathrm{c}_i^j, \mathrm{t}_i^j, \mathrm{ce}_i^j, \mathrm{perf}_i^j, \mathrm{risk}_i^j\}$.

B. Product Design Change Propagation Model under Uncertainty

When an initial change occurs in a part of the product, it will have a certain impact on its related parts, called this phenomenon as design change propagation [6]. In order to determine the scope of affected parts and the affected strength of each part, it is necessary to establish an appropriate product design change propagation model.

The change degree of a node typically varies over time, the higher the change degree of a node, the greater its impact on the service performance of the product. For example, if a changed part has a replacement with inferior performance, i.e., the replacement's performance parameters are lower than the changed part's parameters, the product change intensity will be greater. On the other hand, if the changed part has a replacement with better performance, the product change intensity will be comparatively smaller. Specifically, when a part is influenced by the change effect of its upstream parts, the change intensity of the downstream part p_i can be calculated based on the product network defined in Section II-A, as follows:

$$\bar{r}(p_i) = \sum_{j \in us(i)} \bar{w}_{ij} \times \bar{r}(p_j) \tag{3}$$

where $us(i)$ is the set of nodes upstream of the part p_i. For an upstream part p_j in $us(i)$, $\bar{w}_{ij}$ is the associated weight between the parts p_i and p_j, $r(p_j)$ is the change intensity of the part p_j [27]. It is worth noting that, since the associated weights obtained in this paper are interval numbers, the change

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intensity obtained here is also an interval value, so the interval multiplication elaborated in [29] is used in Formula 3.

When a change undergoes, the impact intensity of the change on each related part along the propagation path is calculated by using the above formula until the initial change intensity is fully absorbed. Considering the weights introduced in the model are interval numbers, the change propagation termination condition is set as follows:

$$P(\bar{r}(p_i) \leq \varepsilon) > \alpha \tag{4}$$

where $P(\bar{r}(p_i) \leq \varepsilon)$ is the probability that the change intensity of part p_i is less than ε, and $\alpha \in [0.95, 01.0]$ represents the confidence level. The parameter ε generally assumes a small value, and following the recommendation of literature [18], in the paper ε is set to 0.001. When the probability that the change intensity of a part exceeds α is less than ε, it can be considered that the current part has fully absorbed the initial change intensity, and the change ends, the impact degrees of parts are denoted as:

$$R(P) = \{\bar{r}(p_1), \bar{r}(p_2), \ldots, \bar{r}(p_n)\} \tag{5}$$

In particular, if $\bar{r}(p_i) = 0$, it signifies that the part p_i is unaffected by the change. For ease of use, this paper stores the parts affected by the change (i.e., $P(\bar{r}(p_i) \leq \varepsilon_i) > \alpha$) in the set $IFP = \{IFp_1, IFp_2, \ldots, IFp_i, \ldots, IFp_{n''}\}$, where n'' means the length of the change path.

III. Interval Multi-objective Optimization Model for the Problem

This section introduces new indicators such as service performance impact degree, carbon cost, and supply risk, to evaluate a design change plan comprehensively, besides the two commonly used indicators, i.e., production cost and duration. Based on these indicators, the problem of product design change plan generation is modeled as a multi-objective combinatorial optimization problem. First, the relevant assumptions required for the proposed model are given.

A. Model Assumptions

(1) Compared to the carbon emissions generated during the actual production and assembly of the product, the carbon emissions during research and development, planning, and small-scale trial production phases are negligible.

(2) Each part of a product has only one primary supplier. When the supplier or the part changes, a new primary supplier is selected from those alternative suppliers as needed.

(3) For a part affected by the change, if the product manufacturer decides to re-develop it, it is assumed that the newly developed part has similar performance to the old one. However, if they decide to replace it with an alternative part from other suppliers, the parameter performance of the new part must be re-evaluated.

(4) Change propagation occurs only when two parts have a connection.

(5) Each edge in the product change network propagates only once when there is a product change. In other words, there are no situations where the same part is changed multiple times or the change effects cyclically propagate along a single path.

B. Representation of a Design Change Plan

For a part p_i affected by a change, the following three change response strategies can usually be adopted.

Strategy 1. Re-developing the part. For the part p_i to be changed, this strategy needs to consider factors such as re-developing cost and duration. In this paper, the total re-developing cost and duration of the changed part are denoted as c_i^{rdev} and t_i^{rdev}, respectively. Actually, a part often only needs to be partially re-developed based on the impact degree of change, and the re-developed part meets the performance requirements of the product. More importantly, those parts re-developed by enterprises themselves are less constrained by suppliers, and their production costs and durations are relatively controllable.

Strategy 2. Replacing p_i by a new part from alternative suppliers. For the part p_i to be changed, this strategy replaces p_i by a new part p_i^j from its alternative part pool. Generally, different alternative parts have different levels of attribute values, including parameter performance $perf_i^j$, carbon emission ce_i^j, manufacturing cost c_i^j, production duration t_i^j, and supply risk $risk_i^j$.

Strategy 3. Do not make any changes. For the part p_i to be changed, decision-makers may choose not to change it due to supplier or cost constraints. In this case, no change duration and cost are incurred, but the product service performance may decline.

C. Decision Variables and Encoding

For a product design change plan (path), it is necessary to consider the parts covered by the path and their adopted change response strategies. To this end, this paper divides the encoding of a change plan into two parts: path node encoding and response strategy encoding. First, the path nodes are encoded by using integers. Taking the three change propagation paths shown in Figure 2 as an example, their encoding results are as follows: 1-3-7-9-12-17, 1-3-7-9-11-15-23, and 1-5-6.

Second, the change response strategies adopted by parts on the change path are encoded as follows. For an affected part p_i, if it chooses not to make changes (i.e., Strategy 3), its value is set to be $\varphi_i = 0$; if it selects Strategy 1 for re-development, its value is set to be $\varphi_i = -1$; if the part chooses a new supplier, that is, the Strategy 2, then $\varphi_i = a_i \in 1, 2, 3, \cdots, |s_i|$, where a_i represents the $a_i - th$ alternative supplier of the part. Furthermore, using an odd-even crossover method to represent the affected change parts and their response strategies, a complete candidate solution of the problem (i.e., a product design change plan) can be given as follows: $DV = \{p_1, \varphi_1, \ldots, p_i, \varphi i, \ldots pn'', \varphi_{n''}\}$, where odd positions represent the order number of part in the change path, even positions represent the change strategy selected by a part, and n'' is the length of the change path.

Taking a change path 1-3-7-9-12-17 in Figure 2 as an example (marked in red), its corresponding encoding result is (1,1,3,0,7,1,9,0,12,-1,17,3), meaning that part 1 selects its first alternative supplier, the part 3 does not change, part 7 selects its first alternative supplier, the part 9 does not change,

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part 12 is re-developed, and part 17 selects the third alternative supplier.

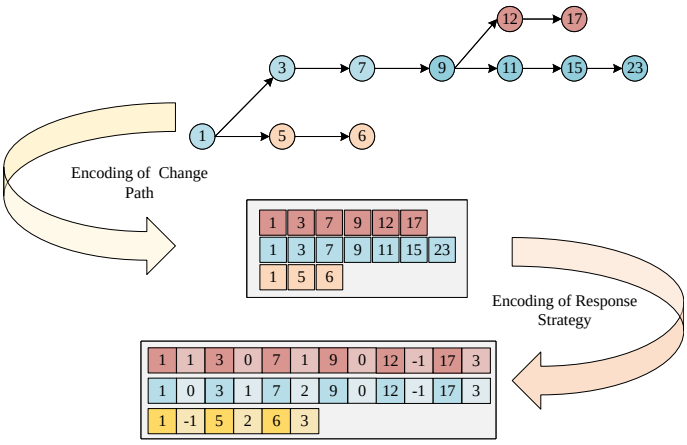


Fig. 2. Encoding of Change Plan

D. Objective Functions

This section considers three optimization objectives: the product service performance impact, the cost of product change taking into account carbon emission trading and supply risk, and the duration of product change

1) Product Service Performance Impact: Changes typically impact the service performance of the product, as some new parts may not provide the same performance parameters as their original parts. The paper evaluates the impact of a product change plan on product service performance by the following formula.

$$\Delta S\bar{P} = \sum_{k=1}^{l} \hat{w}(sp_k) \times \Delta s\bar{p}_k \tag{6}$$

where $\hat{w}(sp_k)$ denotes the weight assigned by the decision-maker to the $k-th$ service performance sp_k, $\Delta s\bar{p}_k$ signifies the impact of the change plan on the $k-th$ service performance. The impact degree of a change plan on sp_k is mainly determined by the degree of impact on the relevant parts. Generally, multiple parts may simultaneously impact a service performance, resulting in a coupling relationship between them. To address the coupling relationship, the node modularity index Q [30] is utilized. For the $k-th$ service performance sp_k, the impact of the change plan can be expressed as Δsp_k:

$$\Delta s\bar{p}_k = \frac{1}{(1+Q_k)} \sum_{p_i,\varphi_i \in DV} i\bar{w}_{ik} \times \Delta \bar{p}erf_i \tag{7}$$

where, Q_k is the modularity value of associated parts, $i\bar{w}_{ik}$ is the impact weight of part p_i on the service performance sp_k. Δperf_i represents the change value in the parameter performance of the part p_i before and after the change, as follows:

$$\Delta \bar{p}erf_i = \begin{cases} 0 & \varphi_i \in -1 \\ \text{perf}_i^0 \times \bar{r}(p_i) & \varphi_i = 0 \\ \text{perf}_i^{\varphi_i} - \text{perf}_i^0 & \varphi_i \in \{1,2,\cdots,|s_i|\} \end{cases} \tag{8}$$

where, $|s_i|$ is the number of alternative suppliers for the part p_i. (1) When the product manufacturer chooses to redevelop the affected part p_i, i.e., $\varphi_i = -1$, the parameter performance of the part is not affected, i.e., $\Delta\text{perf}_i = 0$; (2) when the part remains unchanged, i.e., $\varphi_i = 0$, the value of Δperf_i is primarily determined by its initial parameter performance perf_i^0 and the extent affected by changes $\bar{r}(p_i)$; (3) when the part is replaced by a spare part from another supplier, the value of Δperf_i is the difference in parameter performance between the part and its spare part, i.e., $\text{perf}_i^{\varphi_i} - \text{perf}_i^0$.

2) Change Cost Considering Carbon Emissions and Supply Risk: The traditional product change cost includes two parts: the R&D cost from product redesign, and the new production cost after the change. In order to comprehensively evaluate the cost of product change, this paper introduces carbon emission trading cost and supply risk cost, besides the above traditional product change cost. Carbon emissions trading is currently an important economic means to protect the environment and reduce carbon emissions. Supply risk cost represents the expected cost of supplier risk in the future. Considering the three types of costs, the total change cost of a design change plan is:

$$\Delta\bar{C} = \Delta\bar{C}_1 + \eta\Delta C_2 + \xi\Delta C_3 \tag{9}$$

where, η is the coefficient of carbon emission trading price, ξ is the coefficient of supply risk price. ΔC_1, ΔC_2, and ΔC_3 denote the increment values in economic cost, carbon emission, and supply risk before and after the product change, respectively. According to the literature [31], the range for η is from 0.000053 to 0.000099 (CNY/g), and the value of ξ depends on the total cost of the product. In this research, η is taken as 0.000075 and ξ is taken as 145.67 (10000CNY). Their calculations are as follows.

a) The economic cost of a design change plan consists mainly of the economic cost of each part:

$$\Delta\bar{C}_1 = \sum_{\varphi_i \in DV} \Delta\bar{c}_i \tag{10}$$

where, Δc_i represents the increment value in economic cost when the part p_i is changed. Its calculation method is as follows:

$$\Delta\bar{c}_i = \begin{cases} c_i^{rdev} \times \bar{r}(p_i)/X & \varphi_i = -1 \\ 0 & \varphi_i = 0 \\ c_i^{\varphi_i} - c_i^0 & \varphi_i \in \{1,2,\ldots,|s_i|\} \end{cases} \tag{11}$$

in the above equation, (1) When the product manufacturer chooses to redevelop the affected part p_i, i.e., $\varphi_i = -1$, the increment value in economic cost is decided by the complete redevelopment cost c_i^{rdev}, the change impact degree of the part $r(p_i)$, and the production volume X; (2) when the part remains unchanged, i.e., $\varphi_i = 0$, the increment value Δc_i is 0. (3) When the part is replaced by a spare part from another supplier, the increment value is the difference in economic cost between the part and its spare part $c_i^{\varphi_i} - c_i^0$.

b) The carbon emission and supply risk of a design change plan is also determined by that of each part respectively:

$$\Delta C_2 = \sum_{\varphi_i \in DV} \Delta ce_i \tag{12}$$

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$$\Delta C_3 = \sum_{\varphi_i \in DV} \Delta risk_i \tag{13}$$

where, Δce_i and $\Delta risk_i$ represent the incremental values in carbon emission and supply risk when the part p_i is changed, respectively, as follows:

$$\Delta \text{ce}_i = \begin{cases} 0 & \varphi_i = \{-1, 0\} \\ ce_i^{\varphi_i} - ce_i^0 & \varphi_i \in \{1, 2, \ldots, |s_i|\} \end{cases} \tag{14}$$

$$\Delta risk_i = \begin{cases} 0 & \varphi_i \in \{-1, 0\} \\ risk_i^{\varphi_i} - risk_i^0 & \varphi_i \in \{1, 2, \ldots, |s_i|\} \end{cases} \tag{15}$$

In the above equation, (1) when the product manufacturer chooses to redevelop the affected part p_i, i.e., $\varphi_i = -1$, we believe that the increment values in both carbon emission and supply risk are small and can be ignored because the manufacturer can use some new technology to control carbon emission and supply risk; (2) when the affected part p_i remains unchanged, its carbon emission and supply risk also remain unchanged; (3) when the part is replaced by a spare part from another supplier, the increment value in carbon emission (supply risk) is the difference between the part and its spare part, i.e. $ce_i^{\varphi_i} - ce_i^0$ ($risk_i^{\varphi_i} - risk_i^0$).

3) Product Change Duration: The product change duration refers to the total time spent implementing a change plan, and its value can be determined by the change duration of all affected nodes, as follows:

$$\Delta \bar{T} = \sum_{p_i \in DV} \Delta \bar{t}_i \tag{16}$$

When responding to changes, the product's delay duration mainly includes two parts: one is the duration for redesigning parts, and another is the time required to re-procure spare parts. In particular, the change duration of the affected part p_i is calculated as follows:

$$\Delta \text{t}_i = \begin{cases} t_i^{rdev} \times \bar{r}(p_i) & \varphi_i = -1 \\ 0 & \varphi_i = 0 \\ t_i^{\varphi_i} - t_i^0 & \varphi_i \in \{1, 2, \ldots, |s_i|\} \end{cases} \tag{17}$$

in the equation (17), (1) when the product manufacturer chooses to redevelop the affected part p_i, i.e., $\varphi_i = -1$, its duration mainly depends on its complete re-development duration t_i^{rdev} and change impact degree $r(p_i)$; (2) when the affected part p_i remains unchanged, i.e., $\varphi_i = 0$, no additional duration is required; (3) when the part is replaced by a spare part from another supplier, its change duration is the difference in delivery time between the part and its spare part, i.e., $t_i^{\varphi_i} - t_i^0$.

E. Constraints

A feasible change plan must satisfy actual constraints. This paper considers the following basic constraints:

Constraint 1 indicates that in the obtained change plan, the weight $w_{i(i-1)}$ between two adjacent parts p_i, p_{i-1} is not zero, that is, they must a relationship. This constraint can ensure the continuity of product change propagation.

Constraint 2 limits the value range of decision variable φ_i, φ_i is an integer, where $\varphi_i = -1, 0, a(a > 0)$ represent that the change response strategies 1, 3, and 2 are selected, respectively.

Constraint 3 ensures that the value of decision variable φ_i can only take 0 or -1 when there are no alternative parts to choose from, i.e., $|s_i|$ is 0.

Constraint 4 limits that DV can only contain impacted parts whose change propagation intensity is greater than ε. It indirectly provides the termination condition for the change.

Constraint 5 ensures that the reduction degree in a service performance after the change must be within the allowable range, where $\Delta sp_{k\max}$ represents the maximum allowable reduction value in the $k-th$ service performance.

Among these constraints, Constraints 1-3 do not contain uncertain information, and Constraints 4-5 contain interval numbers and are interval constraints. These constraints are specifically expressed as follows:

$$\textbf{Constraint 1 :} \forall i, p_i, p_{i-1} \in DV,\, w_{i(i-1)} \neq 0 \tag{18}$$

$$\textbf{Constraint 2 :} \forall \varphi_i \in DV, \varphi_i \in \text{Z}, \varphi_i \in [-1, |s_i|], \quad i = 1, 2, \cdots, n'' \tag{19}$$

$$\textbf{Constraint 3 :If } |s_i| = 0, \textbf{ Then } \varphi_i \leq 0, \quad i = 1, 2, \cdots, n''. \tag{20}$$

$$\textbf{Constraint 4 :} \forall p_i \in DV, \bar{r}(p_i) > \varepsilon \tag{21}$$

$$\textbf{Constraint 5 :} \Delta \bar{sp}_k \leq \Delta sp_{k\max}, k = 1, 2, \cdots m. \tag{22}$$

F. Multi-objective Optimization Model of the Problem

Combining the above objective functions and constraints, the problem of product change plan generation with carbon emission and change risk is modeled as the following interval multi-objective optimization model:

$$\begin{aligned} &F(DV) = \min(\Delta \bar{SP}, \Delta \bar{C}, \Delta \bar{T}) \\ &\text{s.t.Constraints}(18) - (22) \end{aligned} \tag{23}$$

In real-world product design change problems, the three objectives usually conflict with each other. To reduce the product change duration, it is necessary to choose suppliers with relatively low risk, resulting in increased costs. To ensure that the product service performance is minimally affected by changes, it is necessary to redevelop more parts, thereby increasing the duration and cost.

Since the associated weights between parts, as well as between parts and service performance have interval uncertainty, the change intensity of part related to it is also an interval number. Furthermore, as the objective functions such as service performance, change cost, and change duration are related to the change intensity of part, their values are also interval numbers. Additionally, Constraints 4 and 5 are also characterized by interval uncertainty.

IV. The Proposed Interval Multi-objective Evolutionary Generation Algorithm

To address the interval multi-objective optimization model in Section III, we propose an enhanced NSGA-II algorithm

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based on interval dominance relationship, called IP-NSGA-II. Several novel operators are developed to address the challenges posed by interval uncertainty. First, the characteristics of the encoding are analyzed, and a problem characteristic-guided population update strategy is devised. Second, a probability-based interval Pareto dominance criterion is introduced to compare individuals with interval objective values. Finally, a novel constraint-handling strategy is proposed for both uncertain and precise constraints.

A. The Algorithm Framework

This section presents the framework of the proposed IP-NSGA-II algorithm for generating product design change plans. Compared to NSGA-II, IP-NSGA-II introduces three improvement strategies: a problem characteristic-guided population update strategy, an interval Pareto dominance-based individual ranking strategy, and an interval constraint handling strategy. The new population update strategy redefines crossover and mutation by fully integrating problem characteristics. The primary function of the interval Pareto dominance-based individual ranking strategy is to accurately compare individuals with interval objective values, and perform non-dominated sorting on candidate solutions obtained by the algorithm. The interval constraint handling strategy manages infeasible solutions generated during the iteration process, ensuring the feasibility of change plans obtained.

Algorithm 1 shows the pseudocode of IP-NSGA-II. Firstly, the population $P(0)$ is initialized (Line 1 of the algorithm). In this step, the breadth-first algorithm is first used to generate N_{pop} initial paths, the change response strategy of each part node in these paths is randomly set. Based on these paths and their change response strategies, we can form N_{pop} initial individuals. Note that the constraint handling strategy proposed in Section IV-D should also be executed if an initial individual does not satisfy the constraints. After that, the main steps of IP-NSGA-II (Lines 3-11) are executed in a loop until the termination condition is met. In Line 4, IP-NSGA-II uses the redefined crossover and mutation operators to generate new individuals; In Line 5, the proposed interval constraint handling strategies are used to deal with newly generated infeasible individuals; following that, the interval Pareto dominance criterion is adopted to select good individuals for parent population in Line 8.

Algorithm 1	**Basic Steps of IP-NSGA-II**
Input:	Population size N_{pop}, maximum iteration times $iter_{\max}$, crossover probability p_c, mutation probability p_m;
Output:	The set of optimal product change plans.
Step 1:	Initialize the population $P(0)$: Generate N_{pop} initial paths using the breadth-first algorithm
Step 2:	$iter$=0;
Step 3:	**While** $iter < iter_{\max}$, **do**
Step 4:	Perform the crossover and mutation operators given in Section IV-B to generate offspring populations, $Q(t+1)$;
Step 5:	Execute the constraint handling strategy proposed in Section IV-D to repair infeasible solutions;
Step 6:	Calculate the objective function values of new individuals using Equation (23);
Step 7:	Merge parent and offspring populations $P(t)+Q(t+1) \rightarrow R(t+1)$;
Step 8:	Calculate the dominance ranks of individuals in $R(t+1)$, using the interval Pareto dominance criterion proposed in Section IV-C;
Step 9:	Select the best N_{pop} individuals from $R(t+1)$ to form $P(t+1)$ using the dominance rank and the crowding distance indicator;
Step 10:	$iter = iter + 1$;
Step 11:	**End While**
Step 12:	Output the optimal solutions in $P(t)$.

B. Problem Characteristic-guided Population Update Strategy

According to the characteristics of the product design change plan generation problem, the traditional crossover and mutation operators are improved.

1) Crossover Operator: When two parent individuals (representing two change paths) share the same part at a specific position, the position is selected to perform the crossover operator. If the two change paths contain multiple identical parts, one of these parts is randomly selected as the crossover position. For instance, in Figure 3, parts 7 and 12 appear in both paths; thus, one of them is randomly chosen as the crossover position., Without loss of generality, the figure illustrates the specific execution process of the crossover when part 7 is used as the crossover position. During the crossover, the two individuals are split at the designated position by using the selected identical part as the crossover position and then their right halves are exchanged to produce new individuals. Noted that, if the current two parent individuals do not share the same part, new parent individuals are reselected for the crossover.

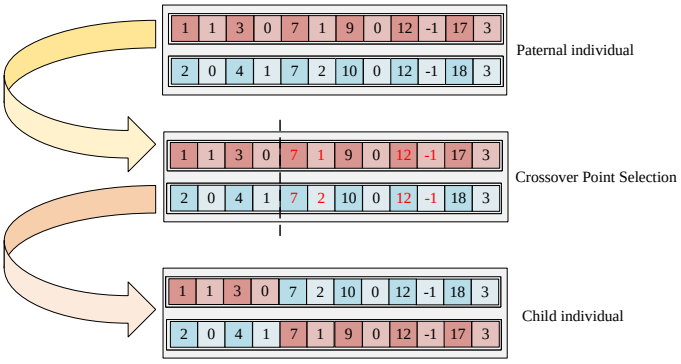


Fig. 3. The Proposed Crossover Operator

2) Mutation Operator: Given the specificity of product change propagation plans, two new mutation operators are redefined: path mutation and strategy mutation. To execute the mutation on the change propagation path, first, a part p_i on the change path is randomly selected as the mutation position, and then its upstream and downstream nodes p_{i-1} and p_{i+1} on the path are determined. Second, the set of all downstream nodes of p_{i-1}, denoted by $Down(p_{i-1})$, and the set of all upstream nodes of p_{i+1}, denoted by $Up(p_{i+1})$ are formed. Following that, a part is randomly select from $Down(p_{i-1}) \cap Up(p_{i+1})$ to replace the part p_i. To perform the mutation on the response strategies, first, a part p_i is randomly selected as the mutation position, and then the change response strategy of this part is

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randomly initialized within the value range.

Taking the product change plan illustrated in Figure 4 as an example, i.e., (1, -1, 3, 1, 7, 2, 9, 0, 12, -1, 17, 3), supposing that the path mutation position selected is the part 3. On the above path, the upstream and downstream nodes of part 3 are parts 1 and 7, respectively. In the whole product network, the downstream node set of part 1 is 3, 4, 5, and the upstream node set of part 7 is also 3, 4, 5. Then, a node we randomly choose from 4, 5 to replace part 3. The mutation result in Figure 4 chose part 5. Supposing that part 12 is selected as the mutation position of the strategy mutation, its value will be initialized by a random value within $-1, 0$. In Figure 4, part 12 randomly selects the response strategy $\varphi_{12} = 0$.

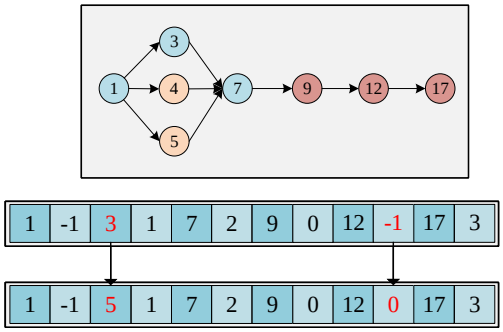


Fig. 4. The Proposed Mutation Operator

C. The Probability-based Interval Pareto Dominance Criterion

The individual evaluation criterion in multi-objective evolutionary algorithms directly impacts the quality of Pareto optimal solutions generated. Generally, researchers use the Pareto dominance relationship to compare different individuals, but this criterion only applies to cases of precise objective values. Evaluation criterion based on interval midpoint or endpoint [32], interval-weighted evaluation method [33], and possibility/similarity-based evaluation criterion [34] are three commonly used methods at present. The first two methods have limitations: they either ignore crucial information, such as interval width, or overly rely on the subjective information of decision-makers, so both cannot accurately compare the size of different interval numbers. The possibility/similarity-based evaluation criterion is recognized by more scholars because it considers both the endpoint of the interval and the uncertainty degree of the interval. Among them, the probability-based Pareto dominance criterion defined in literature [34] is a more typical method. For any two solutions, $\bar{y}$ and $\bar{y}'$, in the M-dimensional objective space, related definitions are as follows.

Definition 1 [34]: In the M-dimensional objective space, $p(\bar{y}' \prec \bar{y}) = \prod_{m=1}^{M} p_m(\bar{y}'_m \prec \bar{y}_m)$ represents the probability that $\bar{y}'$ dominates $\bar{y}$, $p(\bar{y} \prec \bar{y}') = \prod_{m=1}^{M} (1 - p_m(\bar{y}'_m \prec \bar{y}_m))$ represents the probability that $\bar{y}$ dominates $\bar{y}'$, $p(\bar{y} \parallel \bar{y}') = 1 - p(\bar{y}' \prec \bar{y}) - p(\bar{y} \prec \bar{y}')$ is the probability that $\bar{y}$ and $\bar{y}'$ do not dominate each other.

Definition 2 [34]: In the M-dimensional objective space, if the solutions $\bar{y}$ and $\bar{y}'$ satisfy $p(\bar{y}' \prec \bar{y}) = \max\{p(\bar{y}' \prec \bar{y}), p(\bar{y} \prec \bar{y}'), p(\bar{y} \parallel \bar{y}')\}$, then $\bar{y}'$ probabilistically dominates $\bar{y}$, denoted as $\bar{y}' \prec \bar{y}$. At this point, the probability that the decision-maker obtains the best benefit by choosing $\bar{y}'$ is the largest.

In the above formula, $p_m(\bar{y}'_m \prec \bar{y}_m) \in [0, 1]$ represents the probability that the solution $\bar{y}'_m$ is better than $\bar{y}_m$ in the m-th objective space, and its calculation formula is as follows:

$$p_m(\bar{y}'_m \prec \bar{y}_m) = \begin{cases} 0 & {y_m}^+ \le {y'}_m^- \\ \frac{({y_m}^+ - {y'}_m^-)^2}{2w(\bar{y}_m)w(\bar{y}'_m)} & {y_m}^- \le {y'}_m^- < {y_m}^+ \le {y'}_m^+ \\ \frac{2{y_m}^+ - {y'}_m^+ - {y'}^-}{2w(\bar{y}_m)} & {y_m}^- \le {y'}_m^- \le {y'}_m^+ < {y_m}^+ \\ \frac{{y_m}^+ + {y_m}^- - 2{y'}_m^-}{2w(\bar{y}'_m)} & {y'}_m^- \le {y_m}^- \le {y_m}^+ \le {y'}_m^+ \\ 1 - \frac{({y_m}^+ - {y'}_m^-)^2}{2w(\bar{y}_m)w(\bar{y}'_m)} & {y'}_m^- < {y_m}^- \le {y'}_m^+ \le {y_m}^+ \\ 1 & {y'}_m^+ \le {y_m}^- \end{cases} \tag{24}$$

where $w(\bar{y}_m)$ and $w(\bar{y}'_m)$ are the widths of the two interval numbers, it is evident that the larger the value of $p_m(\bar{y}'_m \prec \bar{y}_m)$, the higher the possibility that $\bar{y}'$ has an advantage over $\bar{y}$ in the m-th objective. Specifically, $p_m(\bar{y}'_m \prec \bar{y}_m) = 1$ means that $\bar{y}'$ is entirely superior to $\bar{y}$; $p_m(\bar{y}'_m \prec \bar{y}_m) = 0$ means that $\bar{y}'$ is completely inferior to $\bar{y}$.

However, when dealing with a problem with more objectives, the above probability-based Pareto dominance criterion will retain a lot of mutually non-dominant solutions, leading to a decrease in population selection pressure and seriously affecting the convergence of the algorithm. Taking a 3-objective optimization problem as an example, when $0.7913 > p_1(\bar{y}'_1 \prec \bar{y}_1) = p_2(\bar{y}'_2 \prec \bar{y}_2) = p_3(\bar{y}'_3 \prec \bar{y}_3) > 0.2087$, the criterion all states that $\bar{y}$ and $\bar{y}'$ do not dominate each other. In this case, the criterion will mistakenly judge many solutions as non-dominant solutions and save them. To improve the capability of the above dominance criterion to tackle the above case, this section proposes an improved probability-based interval Pareto dominance criterion.

Definition 3: In an M-dimensional objective space, if $\bar{y}$ and $\bar{y}'$ satisfy the following conditions: (i)$p(\bar{y}'||\bar{y}) = \max\{p(\bar{y}' \prec \bar{y}), p(\bar{y} \prec \bar{y}'), p(\bar{y} \,||\, \bar{y}')\}$; (ii) $[uc(\bar{y}', \bar{y}) > 0] \wedge [\forall m, p_m(\bar{y}'_m \prec \bar{y}_m) \ge p_m(\bar{y}_m \prec \bar{y}'_m)]$, that is, $\bar{y}'$ has a smaller uncertainty than $\bar{y}$ and it is better than $\bar{y}$ in each objective at a higher probability, then $\bar{y}'$ is considered to dominate $\bar{y}$, denoted as $\bar{y}' \prec \bar{y}$.

Correspondingly, if $[uc(\bar{y}', \bar{y}) < 0] \wedge [\forall m, p_m(\bar{y}_m \prec \bar{y}'_m) \ge p_m(\bar{y}'_m \prec \bar{y}_m)]$, then $\bar{y}$ is considered to dominate $\bar{y}'$, denoted as $\bar{y} \prec \bar{y}'$. Otherwise, $\bar{y}'$ and $\bar{y}$ do not dominate each other, denoted as $\bar{y} \,||\, \bar{y}'$.

$$\begin{aligned} uc(\bar{y}', \bar{y}) &= \frac{w(\bar{y}) - w(\bar{y}')}{\max(w(\bar{y}'), w(\bar{y}))} \\ w(\bar{y}) &= \sum_{1}^{m} w(\bar{y}_m) \end{aligned} \tag{25}$$

In the above definition, $uc(\bar{y}', \bar{y})$ is an uncertainty coefficient, as shown in Equation 25. It can be seen that the larger the value of $uc(\bar{y}', \bar{y})$, the smaller the uncertainty of $\bar{y}'$ relative to $\bar{y}$. When $uc(\bar{y}', \bar{y}) > 0$, $\bar{y}'$ has a smaller uncertainty; when $uc(\bar{y}', \bar{y}) < 0$, $\bar{y}$ has a smaller uncertainty.

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Compared with the dominance relationship in reference [34], the improved interval Pareto dominance criterion considers both the magnitude and the uncertainty degree of interval numbers. This can help the algorithm to further screen out pseudo non-dominated solutions, making the population select better offspring individuals. Taking the two solutions $\bar{y} = ([2,4],[2.5,4],[3,4.5])$ and $\bar{y}' = ([3,5],[3,5],[3,5])$ in the three-dimensional objective space as example, we have $p(\bar{y}' \prec \bar{y}) = 0.007$, $p(\bar{y} \prec \bar{y}') = 0.456$, and $p(\bar{y} \,||\, \bar{y}') = 0.537$. The method in reference [34] concludes that the two solutions do not dominate each other. However, the dominance relationship proposed in the paper can distinguish their quality, having the conclusion that $\bar{y}$ is better than $\bar{y}'$.

Other references have also proposed strategies for handling interval objectives. IP-MOEA [35] evaluates the dominance relationship using two endpoints of an interval, but this approach does not fully utilize the characteristics of interval numbers. I-MOMA-II [36] integrates a local search strategy into IP-MOEA, but it only considers interval endpoints. IMOEA/D [37] introduced three new techniques to enhance MOEA/D's ability to handle interval individuals. In the proposed ensemble comparison strategy of interval individuals, multiple sub-strategies are used simultaneously to evaluate interval individuals, but the imprecision of sub-strategies will affect the evaluation results. Moreover, in the proposed adaptive adjusting strategy of reference vector, the setting of the adjustment period and the number of reference vectors will affect the performance of the algorithm. However, IMOEA/D sets these parameters to fixed values, which may limit its ability to solve different problems. The dominance criterion proposed in the paper can reduce the number of mutually non-dominating solutions in the population, thereby accelerating the convergence rate of the algorithm to a certain extent.

D. Interval Constraint Handling Strategy

To ensure the feasibility of new individuals, a reasonable constraint-handling strategy should be designed. The model established in this paper includes precise constraints and interval constraints. First, we convert these interval constraints to precise constraints by introducing a decision-maker tolerance coefficient.

1) Interval Constraint Conversion: The most direct method to deal with constraints is to delete all infeasible individuals generated in an iterative process. Although this method is very simple, it will delete some high-quality individuals with minor constraint violations but close to the true Pareto front. Retaining these high-quality individuals may guide the population to find optimal feasible solutions to the problem quickly. In addition, decision-makers often have a certain tolerance for constraint violation degree in practical product design. In view of this, this section introduces the concept of constraint violation tolerance [38]. When the degree to which a solution satisfies an interval constraint is greater than the tolerance μ_A, the solution is also considered to meet the requirement. Specifically, Constraints 4 and 5 can be re-

Algorithm 2	**The Proposed Infeasible solution repair strategy**
Input:	Infeasible solution, $uDV = (p_1, \varphi_1, \varphi_2, \cdots, p_{n''}, \varphi_{n''})$;
Output:	Repaired feasible solution.
Step 1:	$k = 1$;
Step 2:	**For** $i = 1 : n''$; $\%n''$ is the length of the decision variable
Step 3:	**If** (p_i, φ_i) violates **Constraint 3**;
Step 4:	Randomly set the encoding value of φ_i to 0 or 1;
Step 5:	**EndIf**
Step 6:	**If** (p_i, φ_i) violates **Constraint 4**;
Step 7:	Obtain the parts in φ_i that violate **Constraint 4** and delete them.
Step 8:	**EndIf**
Step 9:	**EndFor**
Step 10:	**If** uDV satisfies **Constraint 5**;
Step 11:	Output the repaired solution.
Step 12:	**Else**
Step 13:	**If** $k \leq \max rptimes$;
Step 14:	$k = k + 1$; **Goto** Step 2.
Step 15:	**Else**
Step 16:	Removal uDV from the population.
Step 17:	**EndIf**
Step 18:	**EndIf**

expressed as follows:

$$\forall p_i \in DV, P(\bar{r}(p_i) > \varepsilon) \geq 1 - \mu_A \tag{26}$$

$$P(\Delta s\bar{p}_k \leq \Delta sp_{k_{\max}}) \geq 1 - \mu_A \tag{27}$$

Further, according to a given constraint violation tolerance coefficient, the above interval constraints can be converted into the following precise constraints:

$$\forall p_i \in DV, r^-(p_i) + \mu_A[r^+(p_i) - r^-(p_i)] > \varepsilon \tag{28}$$

$$\Delta sp^+{}_k - \mu_A[\Delta sp^+{}_k - \Delta sp^-{}_k] \leq \Delta sp_{k_{\max}} \tag{29}$$

where $r^+(p_i)$, $r^-(p_i)$, $\Delta sp^+{}_k$ and $\Delta sp^-{}_k$ are the endpoints of the interval numbers $\bar{r}(p_i)$ and $\Delta s\bar{p}_k$, respectively.

2) Deterministic Constraint Handling Method: Based on the above conversion strategy, all five constraints in the model become precise constraints. This section proposes a repair strategy for infeasible solutions. Since Constraints 1 and 2 can be satisfied directly through the encoding rule, there is no need to deal with these constraints separately. For an infeasible solution that violates **Constraint 3**, we first find the specific parts that violate the constraint and then reset its change response strategy in the space $\{-1, 0\}$. For an infeasible solution that violates **Constraint 4**, those parts that violate **Constraint 4** are directly deleted. For a solution that violates **Constraint 5**, the repair operation for Constraints 3-4 above is re-executed. If the current solution still does not satisfy **Constraint 5** after performing these operations multiple times, we directly delete it from the population. Specifically, Algorithm 2 provides the pseudocode for the repair strategy.

V. Experimental Analysis

This section validates the effectiveness of the proposed IP-NSGA-II algorithm through two sets of experiments. The first set of experiments verifies the effectiveness of the proposed three new strategies. The second set of experiments compares the proposed algorithm with three classical multi-objective evolutionary algorithms, three interval multi-objective evolutionary algorithms, and a change path optimization algorithm.

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A. Cases and Experimental Environment

To verify the effectiveness of the established model and the proposed algorithm, a specific TV model from Skyworth RGB Electronics Co., Ltd. is used. This TV product includes 10 modules, such as a Wi-Fi module, screen module, etc., and 101 different parts. Based on the physical and functional relationships among parts, the DSM (Design Structure Matrix) and multi-layer product network model of the TV product are established, with the DSM shown in Figure 5.

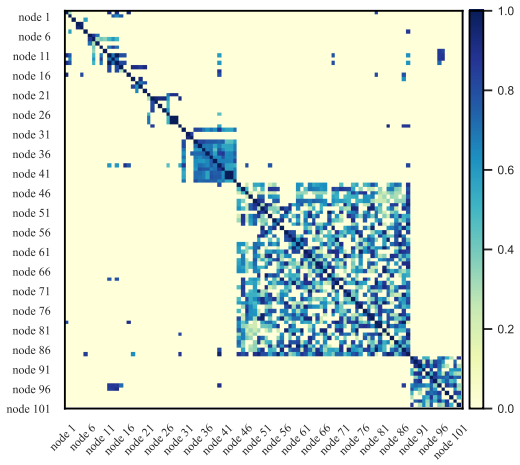


Fig. 5. Design Structure Matrix of a Skyworth TV

According to the company's historical change data, 8 easy-to-change parts are selected as initial change sources. The information of these initial change sources is shown in Table I. For each initial change source in Table I, an interval multi-objective optimization model is established using the method proposed in Section III, denoted the eight optimization models as CPD1-CPD8.

TABLE I
INITIAL CHANGE PARTS OF THE PRODUCT

Problem Name	Initial Part No.	Initial Change Nodes
CPD1	1	Electronic Wire
CPD2	13	Packaged LED
CPD3	26	USB Port
CPD4	28	Crystal Oscillation
CPD5	36	Schottky Diode
CPD6	82	DC-DC module
CPD7	87	Linear Voltage Regulator ICs
CPD8	96	Backlight Driver IC

The Hyper-Volume metric (HV) [39] and ISP metric [34] are used to evaluate optimal solution sets obtained by an algorithm. The HV metric can simultaneously evaluate the diversity and convergence of an optimal solution set. The bigger the HV value of an optimal solution set, the better the quality of the solution set. The ISP metric is mainly used to evaluate the uniform distribution of an optimal solution set in the objective space, with a smaller ISP value indicating a better uniform distribution. For the case that the objective values of optimal solutions are interval, the midpoint of an interval number is used to calculate the HV and ISP values. In the following experiments, the reference point for the HV metric is set to [1,1,1]. Each algorithm is run 30 times. Wilcoxon signed-rank test [40] at 5% significance level is used for statistical analysis, where "+", "=" and "-" represent that the proposed algorithm (strategy) is significantly better than, close to, or significantly worse than the comparison algorithm (strategy).

The objective values of change service performance impact, change cost and change duration are normalized for ease of comparison. To fairly compare the performance of different algorithms, all experiments use the same running environment, i.e., Intel Core(TM) i7 CPU, 32.00GB RAM storage.

B. Analyses on Key Strategies

This section designs three experiments to verify the effectiveness of the proposed problem characteristic-guided population update strategy, the probability-based interval Pareto dominance criterion, and the interval constraint handling strategy. With the exception of the comparative strategy to be analyzed, each experiment uses the same parameter settings. Here, the population size is uniformly set to 200, and the maximal iteration time is uniformly set to 100.

1) The Proposed Population Update Strategy: Section IV-B introduces a problem characteristic-guided population update strategy. To verify its efficiency, this section compares it with the following two crossover and mutation strategies: (a) Simulated binary crossover [41] and polynomial mutation [41] strategy commonly used in NSGA-II algorithm (denoted as SBX+PM); (b) Order crossover [42] and inversion mutation strategy [42] commonly used in path planning optimization algorithms (denoted as OX+INV). For the eight product change problems, Table S1 in **the Supplementary Document** shows the HV and ISP values obtained by the proposed update strategy and the two comparison strategies.

Table S1 shows that: a) for all the eight product design change problems, the proposed population update strategy achieved the best HV and ISP mean values in 8 and 6 problems, respectively; for the remaining problems, the proposed population update strategy also obtained the second-ranked mean values; b) their Wilcoxon test results show that in terms of the HV metric, the proposed population update strategy is significantly better than the two comparison strategies on all eight problems; in terms of the ISP metric, the proposed population update strategy has a significant advantage to the SBX+PM strategy on 6 problems, and has a significant advantage to the OX+INV strategy on 7 problems. Overall, the proposed population update strategy significantly improves the ability of evolutionary algorithms to solve product change problems.

2) The Probability-based Interval Pareto Dominance Criterion: Section IV-C presented an improved probability-based interval Pareto dominance criterion (PIP-DC), this section compares it with the interval dominance criterion (I-DC) proposed in [34] and the direct conversion-based strategy (DCS). In DCS, first, an interval number is converted to a precise value by interval midpoint, and then the traditional Pareto dominance relationship is used. For the eight product

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change problems, Table II shows the HV and ISP values obtained by the proposed dominance criterion and the two existing methods. Table II shows that: a) compared with the direct conversion-based strategy, the proposed PIP-DC and the I-DC in [34] both show better performance in HV and ISP. The main reason is that the direct conversion-based strategy ignores essential information such as interval width; b) Compared with the I-DC, the proposed PIP-DC achieved the best HV and ISP mean values on 8 and 6 problems, respectively. Furthermore, their Wilcoxon test results show that the HV values of the proposed PIP-DC are significantly better than those of I-DC on all 8 problems, and its ISP values are significantly better than those of I-DC on 6 problems. In summary, the proposed PIP-DC is also effective.

TABLE II
HV AND ISP VALUES OBTAINED BY THE PROPOSED PIP-DC AND THE TWO EXISTING METHODS[MEAN(STD)]

Problem	The Proposed Dominance Criteria		Direct Conversion Strategy	
	HV	ISP	HV	ISP
CPD1	**0.916(0.175e-2)**	**0.168(2.647e-2)**	0.867(7.282e-2) +	0.219(8.955e-2) +
CPD2	**0.824(0.620e-2)**	**0.223(0.120e-2)**	0.776(7.126e-2) +	0.244(5.514e-2) +
CPD3	**0.828(0.187e-2)**	**0.156(0.995e-2)**	0.793(7.270e-2) +	0.213(9.037e-2) +
CPD4	**0.860(0.104e-2)**	0.321 (0.448e-2)	0.808(8.667e-2) +	0.336(3.056e-2) +
CPD5	**0.893(0.669e-2)**	**0.106 (4.160e-2)**	0.842(8.692e-2) +	0.212(14.410e-2) +
CPD6	**0.897(0.336e-2)**	0.297(0.408e-2)	0.850(8.909e-2) +	0.316(3.374e-2) +
CPD7	**0.927(0.275e-2)**	**0.164(2.269e-2)**	0.878(10.171e-2) +	0.205(8.095e-2) +
CPD8	**0.916(0.424e-2)**	**0.161(0.690e-2)**	0.874(8.879e-2) +	0.207(8.280e-2) +
+/-/=	/	/	8/0/0	8/0/0

Problem	Dominance Criteria in [34]	
	HV	ISP
CPD1	0.893(1.528e-2) +	0.233(6.453e-2) +
CPD2	0.806(1.265e-2) +	0.276(4.236e-2) +
CPD3	0.812(1.501e-2) +	0.260(4.874e-2) +
CPD4	0.837(1.075e-2) +	**0.283(3.271e-2) -**
CPD5	0.866(1.326e-2) +	0.182(7.247e-2) +
CPD6	0.865(1.972e-2) +	**0.285(3.873e-2) =**
CPD7	0.903(1.657e-2) +	0.269(5.056e-2) +
CPD8	0.899(1.092e-2) +	0.224(4.938e-2) +
+/-/=	8/0/0	6/1/1

3) The Proposed Interval Constraint Handling Strategy: Addressing the interval constraints in the model, Section IV-D proposed an interval constraint handling strategy (IC-HS). The experiment validates the superiority of the proposed IC-HS by comparing it with the method of directly deleting infeasible solutions (denoted by the direct method).

For the 8 product change problems, Table III presents the HV and ISP values obtained by the proposed IC-HS and the direct method. It reports that: a) Compared with the direct method, the proposed IC-HS achieved better HV and ISP values on 8 and 7 problems, respectively; b) Their Wilcoxon test results indicate that the HV values obtained by IC-HS have a significant advantage on all eight problems, and its ISP values are significantly better than the direct method on 6 problems. In summary, the proposed IC-HS effectively enhances the algorithm's ability to solve product change problems.

C. Comparative Experiments

To validate the performance of the proposed IP-NSGA-II algorithm, three classical multi-objective evolutionary algorithms, three interval multi-objective evolutionary algorithms, and a classical change path optimization algorithm are selected for comparison in the experiment.

TABLE III
HV AND ISP VALUES OBTAINED FROM THE PROPOSED IC-HS AND THE DIRECT METHOD [MEAN(STD)]

Problem	The proposed IC-HS		The direct method	
	HV	ISP	HV	ISP
CPD1	**0.916(0.175e-2)**	**0.168(2.647e-2)**	0.886(1.190e-2) +	0.235(6.212e-2) +
CPD2	**0.824(0.620e-2)**	**0.223(0.120e-2)**	0.802(1.191e-2) +	0.292(4.122e-2) +
CPD3	**0.828(0.187e-2)**	**0.156(0.995e-2)**	0.813(1.747e-2) +	0.282(4.906e-2) +
CPD4	**0.860(0.104e-2)**	0.321(0.448e-2)	0.836(1.282e-2) +	**0.289(3.998e-2) -**
CPD5	**0.893(0.669e-2)**	**0.106(4.160e-2)**	0.858(1.715e-2) +	0.207(9.397e-2) +
CPD6	**0.897(0.336e-2)**	**0.297(0.408e-2)**	0.859(1.474e-2) +	**0.297(3.339e-2) =**
CPD7	**0.927(0.275e-2)**	**0.164(2.269e-2)**	0.895(1.361e-2) +	0.267(4.084e-2) +
CPD8	**0.916(0.424e-2)**	**0.161(0.690e-2)**	0.891(1.285e-2) +	0.260(5.696e-2) +
+/-/=	/	/	8/0/0	6/1/1

1) Comparison with Six Multi-objective Evolutionary Algorithms: MOEA/D [43], NSGA-III [44], [45], and UNSGA-III [46] are classical multi-objective evolutionary algorithms for precise objective values. In order to enable these algorithms to handle interval optimization problems, interval midpoints are used directly in the optimization process. Three advanced interval optimization algorithms are IP-MOEA [35], I-MOMA-II [36], and IMOEA/D [37].

To ensure fairness, the size of the population/swarm in these algorithms is uniformly set to be 200, and the maximum iteration time is uniformly set to be 100; the crossover and mutation parameters in NSGA-III are both set to 20; the aggregation method of MOEA/D adopts the boundary intersection method; other parameters of all the comparative algorithms are set to the recommended values given in their original literature. For the eight product change problems, Table IV presents the HV and ISP values obtained by the proposed IP-NSGA-II and the six comparative algorithms.

From the results in Table IV, it can be seen that the proposed IP-NSGA-II achieves the best HV mean values on 6 out of 8 problems and the best ISP mean values on 7 out of problems. Compared with the three multi-objective evolutionary algorithms for precise objectives (MOEA/D, NSGA-III, and UNSGA-III), the four interval algorithms (IP-NSGA-II, IP-MOEA, I-MOMA-II and IMOEA/D) all show relatively good performance. This is mainly attributed to the interval uncertainty processing strategies they adopt. Specifically, we can see that:

a) Compared with the three multi-objective evolutionary algorithms for precise objectives, i.e., MOEA/D, NSGA-III, and UNSGA-III, the HV values of the proposed IP-NSGA-II show significant advantages on all 8 problems. The NSGA-III also obtains relatively good results compared to MOEA/D and U-NSGA-III on most of the problems, because it can efficiently maintain population diversity through a reference point-based selection mechanism [44], [45].

b) Compared with the three interval algorithms, i.e., IP-MOEA, I-MOMA-II and IMOEA/D, the proposed IP-NSGA-II has better HV values on 8, 7 and 7 problems respectively. The ISP values obtained by the proposed algorithm are also significantly better than that of the three comparison algorithms on most of the problems. This may be because the three comparison algorithms all lack effective interval constraint handling strategies, which greatly limits their performance. The proposed IP-NSGA-II can effectively handle both interval

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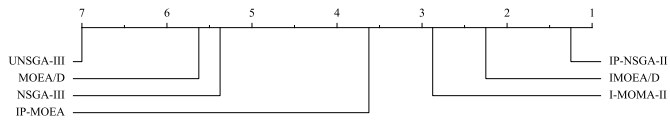
TABLE IV
HV AND ISP VALUES [MEAN(STD)] OBTAINED BY THE PROPOSED ALGORITHM AND THE SIX COMPARISON ALGORITHMS

Problem	MOEA/D HV	MOEA/D ISP	NSGA-III HV	NSGA-III ISP	UNSGA-III HV	UNSGA-III ISP	IP-MOEA HV	IP-MOEA ISP
CPD1	0.834(1.111e-2) +	0.248(6.108e-2) +	0.836(1.246e-2) +	0.249(5.104e-2) +	0.787(1.307e-2) +	0.275(5.837e-2) +	0.882(1.426e-2) +	0.233(6.576e-2) +
CPD2	0.752(1.183e-2) +	0.321(5.229e-2) +	0.751(1.590e-2) +	0.300(4.424e-2) +	0.712(1.387e-2) +	0.316(5.660e-2) +	0.795(0.993e-2) +	0.302(3.022e-2) +
CPD3	0.759(1.185e-2) +	0.258(9.161e-2) +	0.760(1.439e-2) +	0.285(7.319e-2) +	0.727(0.617e-2) +	0.297(8.546e-2) +	0.802(0.672e-2) +	0.252(5.509e-2) +
CPD4	0.781(1.220e-2) +	0.386(4.639e-2) +	0.783(1.560e-2) +	0.309(4.846e-2) =	0.744(1.112e-2) +	0.353(2.915e-2) +	0.829(0.818e-2) +	0.294(5.057e-2) -
CPD5	0.812(1.478e-2) +	0.183(9.596e-2) +	0.804(1.876e-2) +	0.244(9.933e-2) +	0.761(3.836e-2) +	0.184(8.752e-2) +	0.853(1.586e-2) +	0.244(6.713e-2) +
CPD6	0.815(1.630e-2) +	0.353(3.817e-2) +	0.811(1.584e-2) +	0.325(4.099e-2) +	0.776(1.337e-2) +	0.320(3.650e-2) +	0.858(1.369e-2) +	0.305(3.455e-2) +
CPD7	0.839(1.828e-2) +	0.253(7.266e-2) +	0.841(1.603e-2) +	0.270(6.371e-2) +	0.807(1.021e-2) +	0.279(8.335e-2) +	0.878(1.822e-2) +	0.260(5.067e-2) +
CPD8	0.838(0.815e-2) +	0.228(5.649e-2) +	0.841(1.207e-2) +	0.235(5.235e-2) +	0.796(0.947e-2) +	0.250(6.821e-2) +	0.887(1.043e-2) +	0.220(3.987e-2) +
+/-/=	8/0/0	8/0/0	8/0/0	7/0/1	8/0/0	8/0/0	8/0/0	7/1/0

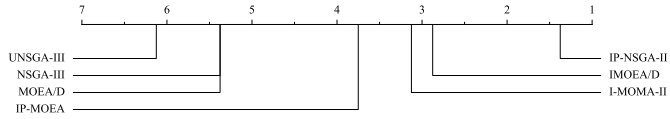
Problem	I-MOMA-II HV	I-MOMA-II ISP	IMOEA/D HV	IMOEA/D ISP	IP-NSGA-II HV	IP-NSGA-II ISP
CPD1	0.883(1.473e-2) +	0.235(5.597e-2) +	0.888(4.194e-2) +	0.194(4.644e-2) +	**0.916(0.175e-2)**	**0.168(2.647e-2)**
CPD2	0.791(1.661e-2) +	0.273(5.305e-2) +	0.800(3.318e-2) +	0.262(5.766e-2) +	**0.824(0.620e-2)**	**0.223(0.120e-2)**
CPD3	0.806(1.284e-2) +	0.254(7.182e-2) +	0.796(3.840e-2) +	0.218(11.74e-2) +	**0.828(0.187e-2)**	**0.156(0.995e-2)**
CPD4	0.828(1.467e-2) +	**0.288(5.720e-2) =**	0.835(3.414e-2) +	0.331(5.027e-2) =	**0.860(0.104e-2)**	0.321(0.448e-2)
CPD5	0.856(1.927e-2) +	0.165(8.875e-2) +	0.863(3.864e-2) +	0.141(9.069e-2) +	**0.893(0.669e-2)**	**0.106(4.160e-2)**
CPD6	0.860(1.608e-2) +	**0.297(2.886e-2) =**	**0.903(4.487e-2) =**	0.329(5.905e-2) +	0.897(0.336e-2)	**0.297(0.408e-2)**
CPD7	0.894(1.615e-2) +	0.231(5.995e-2) +	0.906(3.257e-2) +	0.199(6.841e-2) +	**0.927(0.275e-2)**	**0.164(2.269e-2)**
CPD8	**0.923(1.216e-2) -**	0.230(4.977e-2) +	0.894(3.268e-2) +	0.186(4.001e-2) +	0.916(0.424e-2)	**0.161(0.690e-2)**
+/-/=	7/1/0	6/0/2	7/0/1	7/0/1	/	/

and non-interval constraints, thereby further improving the feasibility of solutions or individuals. Moreover, compared with IMOEA/D, the results of IP-MOEA and I-MOMA-II have poorer results because they only consider interval endpoints.

Furthermore, Figure 6 shows the average ranking results of these algorithms in HV and ISP. In summary, a) all four interval optimization algorithms, including IP-NSGA-II, are superior to the three precise multi-objective algorithms that do not consider interval information; b) the proposed IP-NSGA-II can efficiently handle the eight problems, making it an excellent method for generating product change plans.



(a) Average Ranking of HV Values for 7 Algorithms



(b) Average Ranking of ISP Values for 7 Algorithms

Fig. 6. Average Ranking of Indicator Values for 7 Algorithms

Moreover, to further compare the objective values of the best path change plans obtained from the proposed IP-NSGA-II and other EAs, this paper introduces the fuzzy decision-making method [47] to select an optimal compromise solution from the Pareto optimal solution set obtained by each EA. Taking CPD1 as an example, Table S2 shows the optimal compromise solution obtained by the 8 algorithms and their objective values. It can be seen that the optimal compromise solution of the proposed IP-NSGA-II shows relatively good values on the three objective functions. For the objective function SPI, IP-NSGA-II obtains the minimum objective value; for the change duration, IP-NSGA-II obtains also the second smallest objective value. We also included additional interval results of the proposed IP-NSGA-II and I-MOEA/D algorithms in Table S3-4 in **the Supplementary Document**, which is considered to be the best among six other evolutionary algorithms.

2) Analysis on Two Objectives Sub-Problem: To further demonstrate the effectiveness of the proposed method, the subsection discussed a case with two objectives. Without loss of generality, we set the standard threshold for the service performance impact (SPI) to be 0.5; and transform the objective SPI as a constraint. Consequently, the original optimization model is transformed into a two-objective optimization problem.

Applying the proposed IP-NSGA-II and the three interval EAs (IP-MOEA, I-MOMA-II and IMOEA/D), Table S5 in **the Supplementary Document** shows the HV and ISP values obtained by the four algorithms on the TV model redesign case. We can see from Table S5 that the proposed IP-NSGA-II achieves the best HV and ISP mean values on 7 out of 8 problems. As shown by the Wilcoxon signed-rank test results, the proposed IP-NSGA-II is significantly better than the three comparison algorithms on most problems. In summary, the proposed IP-NSGA-II is also superior to other algorithms in the two objectives sub-problems.

3) Comparison with Classical Change Path Optimization Algorithm: This section compares the proposed IP-NSGA-II with a classical change path optimization algorithm, LRCPP [19]. Without loss of generality, the problem CPD1 is chosen to compare the two algorithms. For the problem CPD1, the LRCPP algorithm provides a relatively optimal change path: part 1 $\rightarrow$ part 14 $\rightarrow$ part 88 $\rightarrow$ part 80 $\rightarrow$ part 85 $\rightarrow$ part 84 $\rightarrow$ part 86. The visual results of these paths were given in **the Supplementary Document** (Figure S1).

To ensure that the LRCPP algorithm can deal with interval numbers in the model, their midpoints are used during the optimization process. For the above change path, using Equation (23) to calculate its three objective values, we have the service performance impact degree=0.54, the change cost=16.33 million CNY, and the change duration=29.06 hours.

Using the proposed IP-NSGA-II to solve the problem CPD1, a set of change plan sets was obtained. Figure 7 shows the Pareto front of the set of change plans, where the pentagram represents the change path obtained by LRCPP, the circles

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TABLE V
CHANGE PLANS OBTAINED BY IP-NSGA-II AND LRCPP

Change Plan No.	Change Path	Response strategy No.	
S0	1,14,88,80,85,84,86	-	
S1	1,80,85,84,86	3,3,0,0,-1	
S2	1,80,85,84,86	4,3,0,0,-1	
S3	1,80,85,84,86	2,3,0,0,-1	
S4	1,80,46,85,84,75,59	1,4,0,0,0,-1,-1	
S5	1,80,49,46,85,84,75	1,2,3,-1,0,0,2	
S6	1,80,47,51,85,84,86	4,0,0,1,0,0,-1	
S7	1,80,49,46,78,85,84	4,2,0,4,4,0,-1	
S8	1,80,49,46,85,84,86	1,0,1,-1,0,0,-1	
Change Plan No.	SPI value	Change Cost	Change Duration
S0	0.54	16.33	29.06
S1	[0.49,0.50]	[11.79,11.85]	[18.68,18.87]
S2	[0.43,0.44]	[12.99,13.05]	[19.01,19.20]
S3	[0.43,0.44]	[13.88,13.94]	[17.39,17.58]
S4	[0.31,0.33]	[23.76,23.77]	[13.89,13.99]
S5	0.20	[28.89,28.95]	[25.61,26.39]
S6	[0.93,0.97]	14.86	8.05
S7	0.20	[29.05,29.06]	[15.81,15.83]
S8	[0.69,0.71]	[4.79,4.87]	[12.50,13.32]

represent the change plan set obtained by the IP-NSGA-II algorithm, the purple circle is a solution which is superior to the solution obtained by LRCPP in the three objectives.

We can see that the change plan obtained by LRCPP is dominated by multiple plans obtained by the IP-NSGA-II. Furthermore, Table V shows part of the change plans obtained by the proposed algorithm. In Table V, S0 is the solution found by LRCPP, S1-S8 are the change plans found by IP-NSGA-II, where $[x^-, x^+]$ represents an interval number, x^- represents the lower limit of interval, and x^+ represents its upper limit of interval. For example, the lower limit of the SPI value for solution S3 is 0.43, its upper limit is 0.44. We also provided the visual results of these paths in **the Supplementary Document** (Figure S2). We can see that IP-NSGA-II can obtain multiple complete change plans, which contain change paths and response strategies, for decision-makers to choose from. However, LRCPP can only obtain one change path and cannot also provide change response strategies. More importantly, multiple change plans(S1,S2,S3) obtained by the proposed algorithm are superior to the change path obtained by LRCPP.

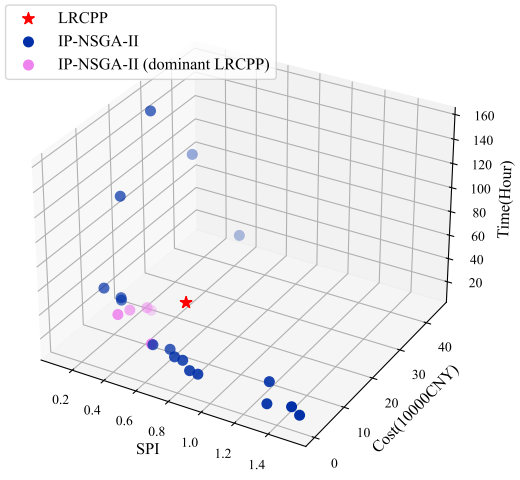


Fig. 7. Pareto Optimal Fronts Obtained by the Two Algorithms

To further compare EAs with the classical change path optimization algorithm, in the experiment we compare the dominance/size relationship between the optimal solutions obtained by EAs and the solution obtained by LRCPP on different objectives. Taking the problem CPD1 as an example, Table S6 in the **Supplementary Document** shows their statistical results (Median/Min/Max). In the table, "Num of solutions" represents the number of optimal solutions obtained by an EA; "Num of better solutions on 3 objectives" represents the number of solutions from an EA that outperform the solution of LRCPP on all the three objectives, "Num of better solutions on 2 objectives" represents the number of solutions from an EA that outperform the solution of LRCPP on any two objectives; "Num of worse solutions" represents the number of solutions from an EA that are inferior to the solution of LRCPP on all the three objectives. It can be seen that all four interval algorithms (i.e., IP-NSGA-II, IP-MOEA, I-MOMA-II, and IMOEA/D) find at least one optimal solution that is better than LRCPP on all three objectives. Most of the optimal solutions from EAs are better than the solution of LRCPP on two objectives. Among them, the IP-NSGA-II performs best, it obtains three optimal solutions that are superior to LRCPP on all three objectives.

4) Computational Complexity: Furthermore, to compare the computational cost of the classical optimization algorithm LRCPP with other EAs, Figure 8 shows the average running times of these algorithms on CPD1-CPD8. It reports that: 1) all the EAs have higher computational cost compared to the classical change path optimization algorithms, but they are able to obtain multiple optimal solutions simultaneously, as shown in Section V.C.3. 2) The four interval EAs show higher computational cost than non-interval EAs; because they require additional computational cost to handle interval numbers. Among the four interval EAs, the proposed IP-NSGA-II has the second-best calculation cost. Overall, considering that the proposed algorithm can obtain a better optimal solution set, its computational cost is also acceptable.

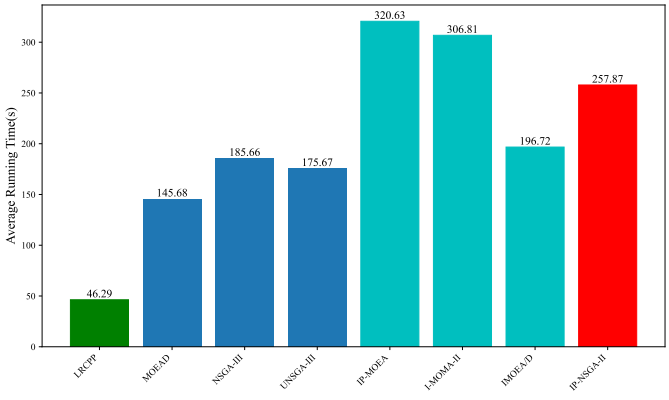


Fig. 8. Average Running Time(s) of the Eight Algorithms on CPD1-CPD8

5) Case Study on Motorcycle Engine Problem: To verify the model's effectiveness and generalizability to different types of products, an open product change model, namely, the design problem of Motorbike engines in [48] is also used in the subsection. Since this original model lacks interval weight data, random interval weights, varying between -5% and +5%, are generated based on the original weights. Additionally,

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since the original dataset lacks relevant information for part redesign, the change response strategy 1 hasn't been considered either in the experiment. Without loss of generality, taking the No.1, No.5, No.10 and No.19 parts as initial change nodes, and both the proposed IP-NSGA-II and other comparative algorithms are used to solve the model. To ensure fairness, the size of the population in these algorithms is uniformly set to 200, and the maximum iteration time is uniformly set to 100. For other key parameters in all the comparison algorithms, the recommended values given in their original literature are used.

For the four product change problems, Table S7 in **the Supplementary File** shows the HV and ISP values obtained by the proposed IP-NSGA-II and three state-of-the-art interval optimization algorithms. From Table S7, it can be concluded that: a) The proposed IP-NSGA-II achieves the best average HV mean values on all 4 problems, and has the best average ISP values on 3 problems; b) In terms of Wilcoxon signed-rank test results, the proposed IP-NSGA-II has better results compared to other comparison algorithms in most cases. This further proves the ability of the proposed algorithm to handle different product design change problems.

VI. Conclusions

This paper proposed an interval multi-objective evolutionary algorithm, called IP-NSGA-II, to find optimal product design change plans under uncertain environments. An interval-based multi-layer network model for products was first developed to tackle the challenge that the degree of correlation between parts is difficult to accurately determine. Following that, the product design change planning problem was modeled as an interval multi-objective optimal model, and some new factors such as product service performance, carbon emissions, and supply risk were introduced into the optimal model. With the help of three new evolutionary operators/strategies, i.e., the problem characteristic-guided population update strategy, the probability-based interval Pareto dominance, and the interval constraint handling strategy, the capability of the proposed IP-NSGA-II was significantly improved. The experimental results on the design problems of TV product and motorcycle engine revealed that IP-NSGA-II is a highly competitive method to deal with the product design change planning problem, it can simultaneously generate a set of optimal design change plans.

Although the proposed IP-NSGA-II has shown good performance in addressing product design change problems, there are still some challenges and limitations. Firstly, the proposed algorithm only considers the case of a single change source, it cannot effectively handle the case with multiple change sources simultaneously. Secondly, the main operators proposed in the paper are for interval uncertainty only, it is still necessary to conduct more in-depth research into other types of uncertainty information. Thirdly, the exploration of adapting other advanced algorithms like MOEA/D to product design change problems appears to be a promising avenue for future research. Moreover, the effectiveness of IP-NSGA-II should be further verified in a broader range of product design change problems in the future.

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